

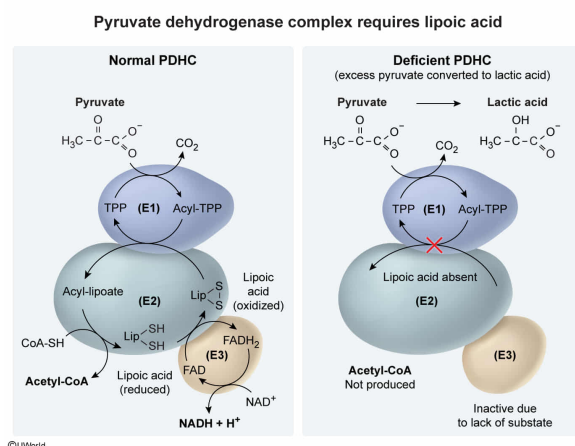
An infant who experiences seizures and has lactic acid buildup is shown to have deficient pyruvate dehydrogenase complex (PDHC) activity. Genetic analysis shows that both the maternal and paternal alleles for all PDHC subunits are normal. These results could indicate:

- ☐ A. overexpression of one or more PDHC subunits. [9%]
- ✓ ☒ B. impaired ability to synthesize lipoic acid. [51%]
- ☐ C. upregulated acetyl-CoA entry into the Krebs cycle. [17%]
- ☐ D. an unusually low rate of pyruvate fermentation. [20%]

Correct

52% answered correctly

Explanation:



The pyruvate dehydrogenase complex (PDHC) is an enzyme composed of three subunits (E1, E2, and E3). It catalyzes the **oxidative decarboxylation** of pyruvate to form acetyl-CoA, along with the reduction of NAD^+ to form NADH. During this process, electrons are passed from one subunit in the complex to the next until they can be transferred to NAD^+ . This transfer is facilitated in part by the **cofactor lipoic acid**.

Cofactors are **non-amino acid chemical compounds** that must bind to certain proteins before those proteins can function. Cofactors can be inorganic (eg, metal ions) or organic (eg, lipoic acid); organic cofactors are sometimes called coenzymes. Because cofactors are not amino acids, they are not directly encoded by DNA.

The question states that PDHC has decreased activity, yet the alleles (DNA) encoding it are normal. Therefore, some component of PDHC activity that is not directly encoded in the DNA must be altered. Because lipoic acid is a necessary cofactor of the PDHC, **impaired lipoic acid production** would result in decreased activity of the PDHC complex.

(Choice A) The question states that alleles for all PDHC subunits are normal.

Therefore, underexpression (not overexpression) would account for decreased activity

of the normal proteins.

(Choice C) The PDHC synthesizes acetyl-CoA, which enters the Krebs cycle (also known as the citric acid cycle). A deficient PDHC would *decrease* acetyl-CoA production and downregulate its entry into the Krebs cycle.

(Choice D) Lactic acid is exclusively produced by pyruvate fermentation, so the fact that the infant experienced lactic acid buildup indicates that the rate of fermentation is high.

Educational objective:

Many enzymes rely on non–amino acid components called cofactors or coenzymes to function. The pyruvate dehydrogenase complex requires lipoic acid for activity.

Time spent:0 sec

Subject:
Biochemistry

QID:402001

System:
Metabolic Reactions